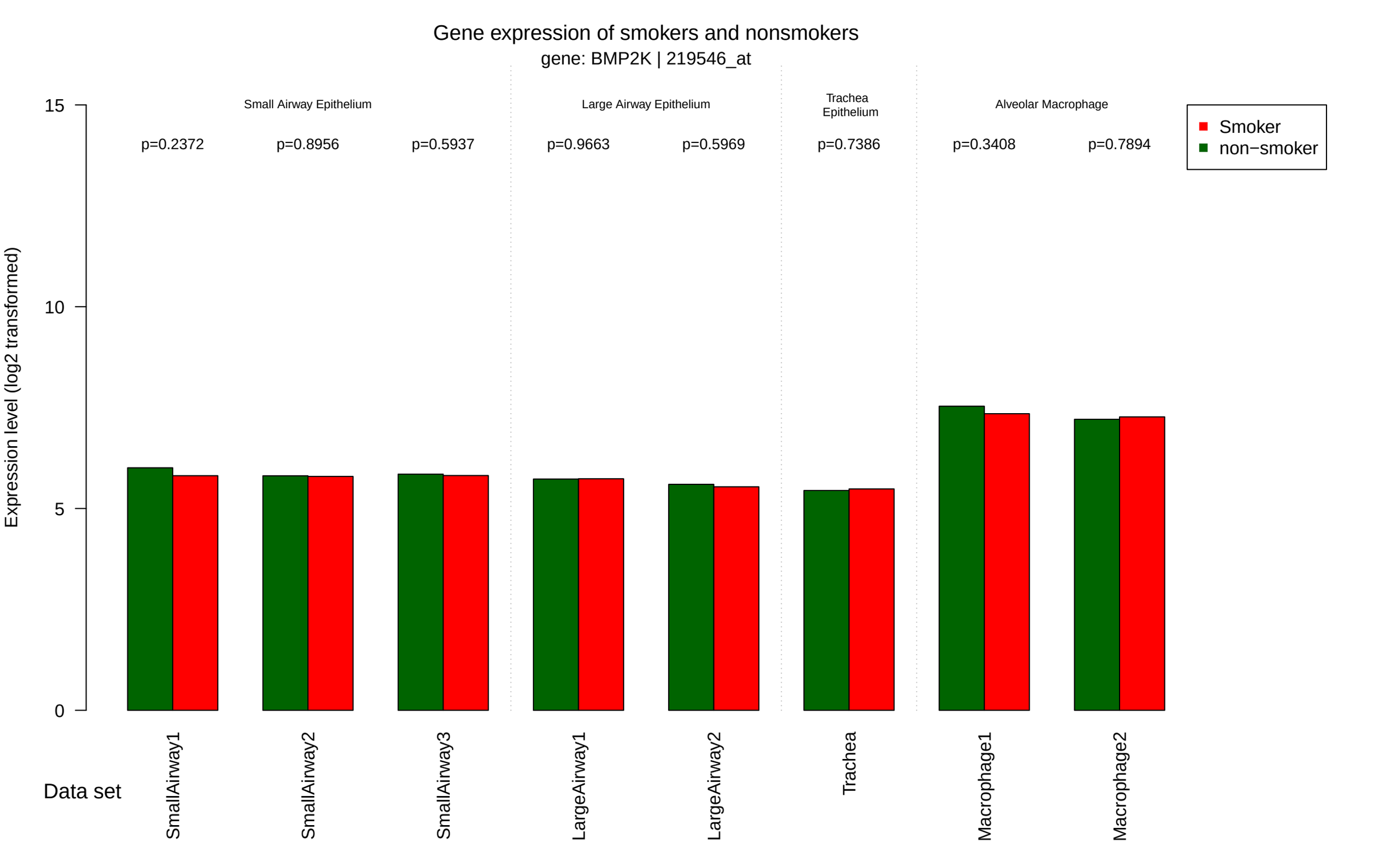
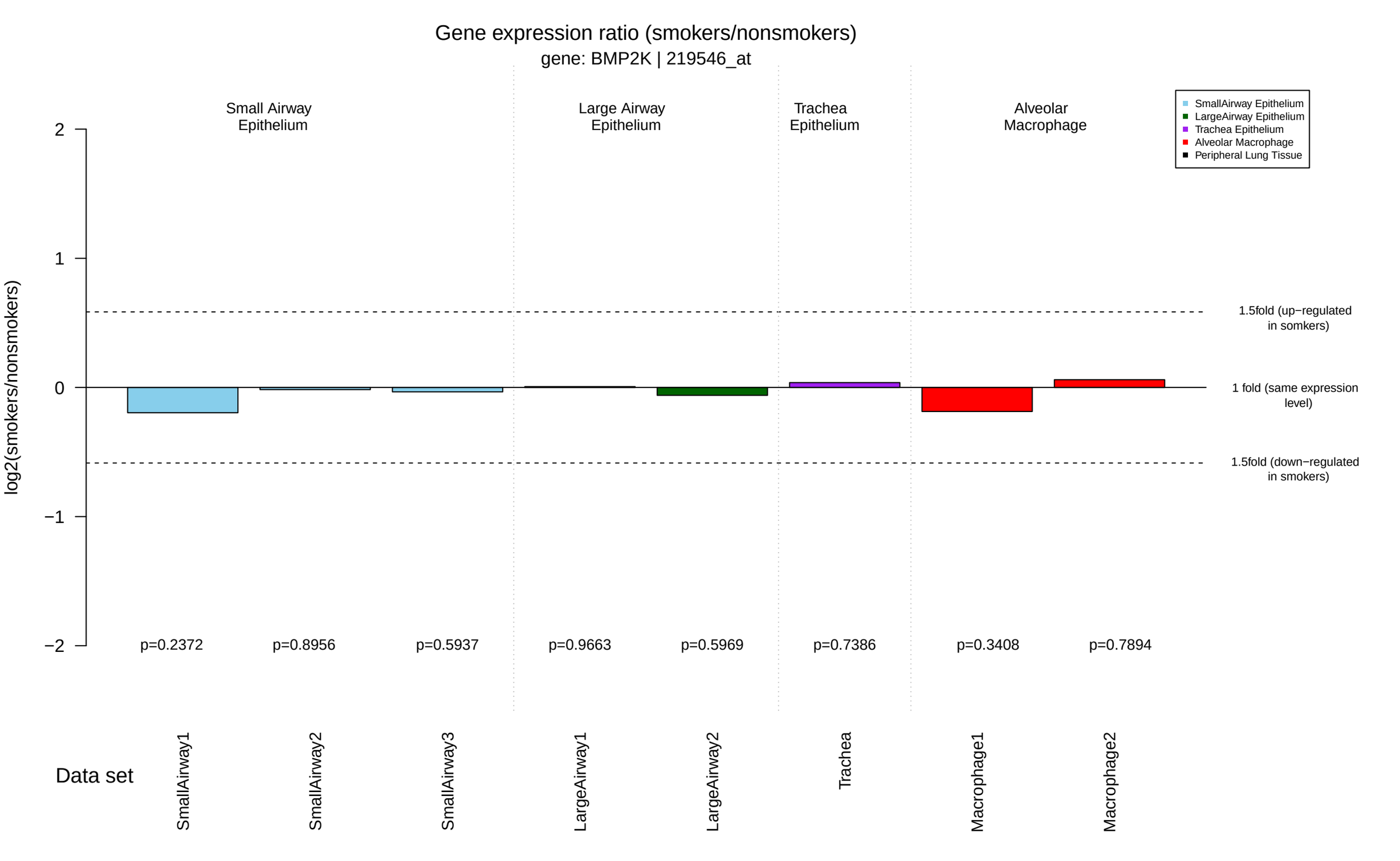
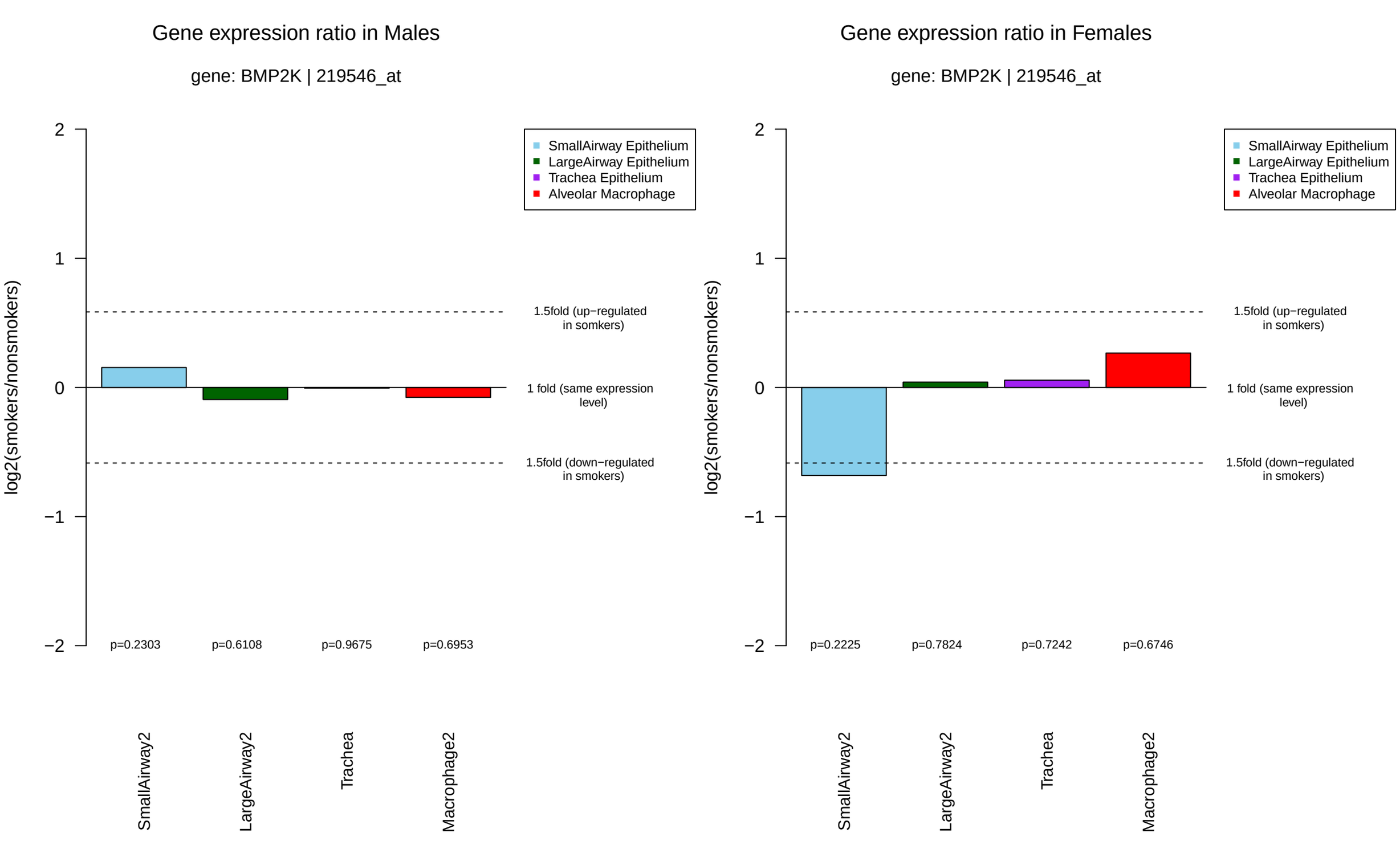
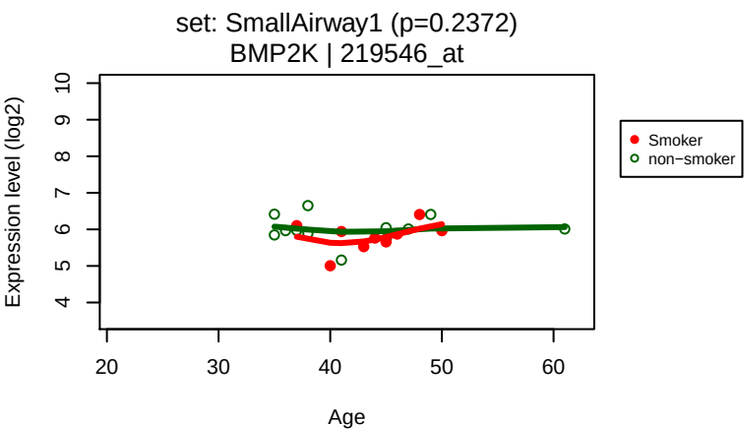
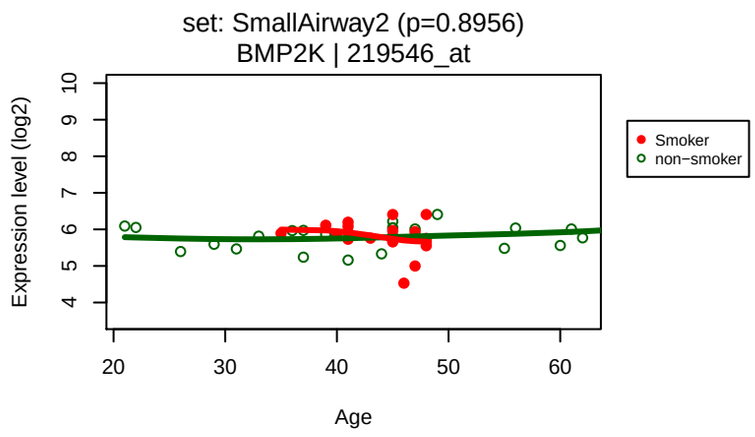
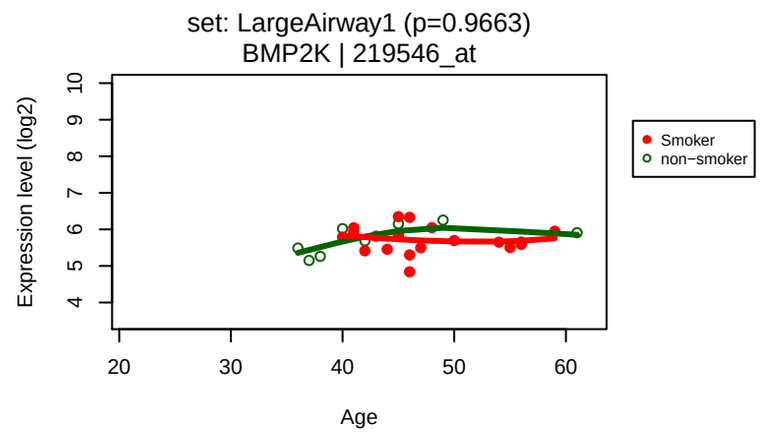

Gene expression of smokers and nonsmokers
gene: BMP2K | 219546_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2372
p=0.8956
p=0.5937
p=0.9663
p=0.5969
p=0.7386
p=0.3408
p=0.7894
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

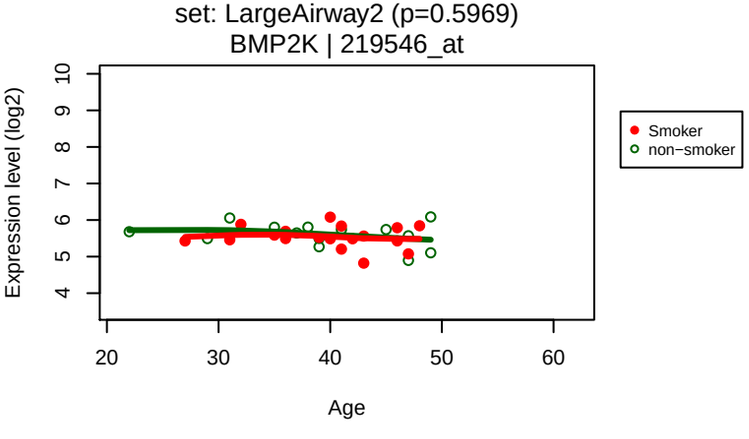

Gene expression ratio (smokers/nonsmokers)
gene: BMP2K | 219546_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2372
p=0.8956
p=0.5937
p=0.9663
p=0.5969
p=0.7386
p=0.3408
p=0.7894
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

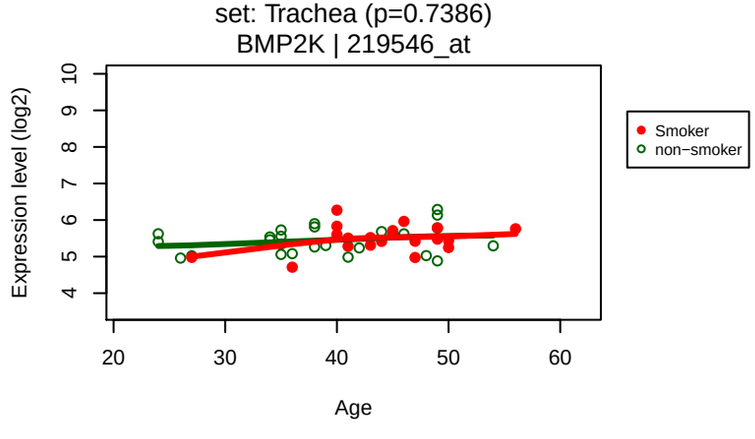

Gene expression ratio in Males
gene: BMP2K | 219546_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2303
p=0.6108
p=0.9675
p=0.6953
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BMP2K | 219546_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2225
p=0.7824
p=0.7242
p=0.6746
SmallAirway2
LargeAirway2
Trachea
Macrophage2

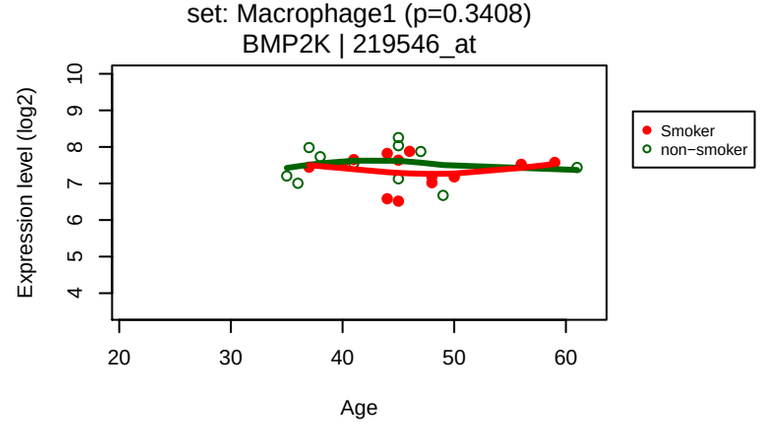

set: SmallAirway1 (p=0.2372)
BMP2K | 219546_at
Expression level (log2)
Age
Smoker
non−smoker

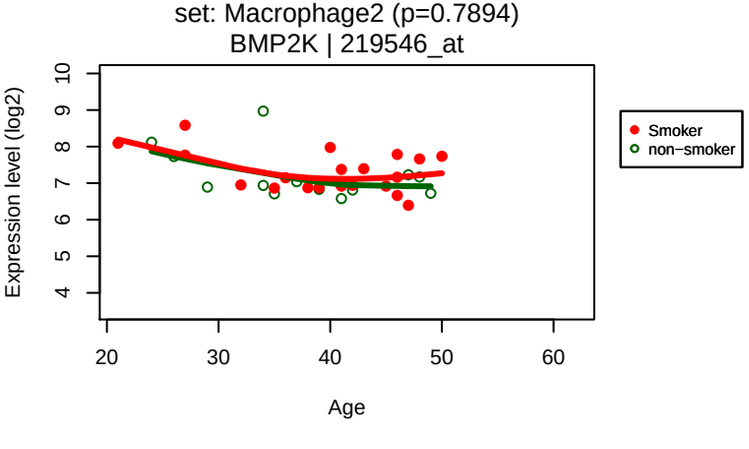

set: SmallAirway2 (p=0.8956)
BMP2K | 219546_at
Expression level (log2)
Age
Smoker
non−smoker

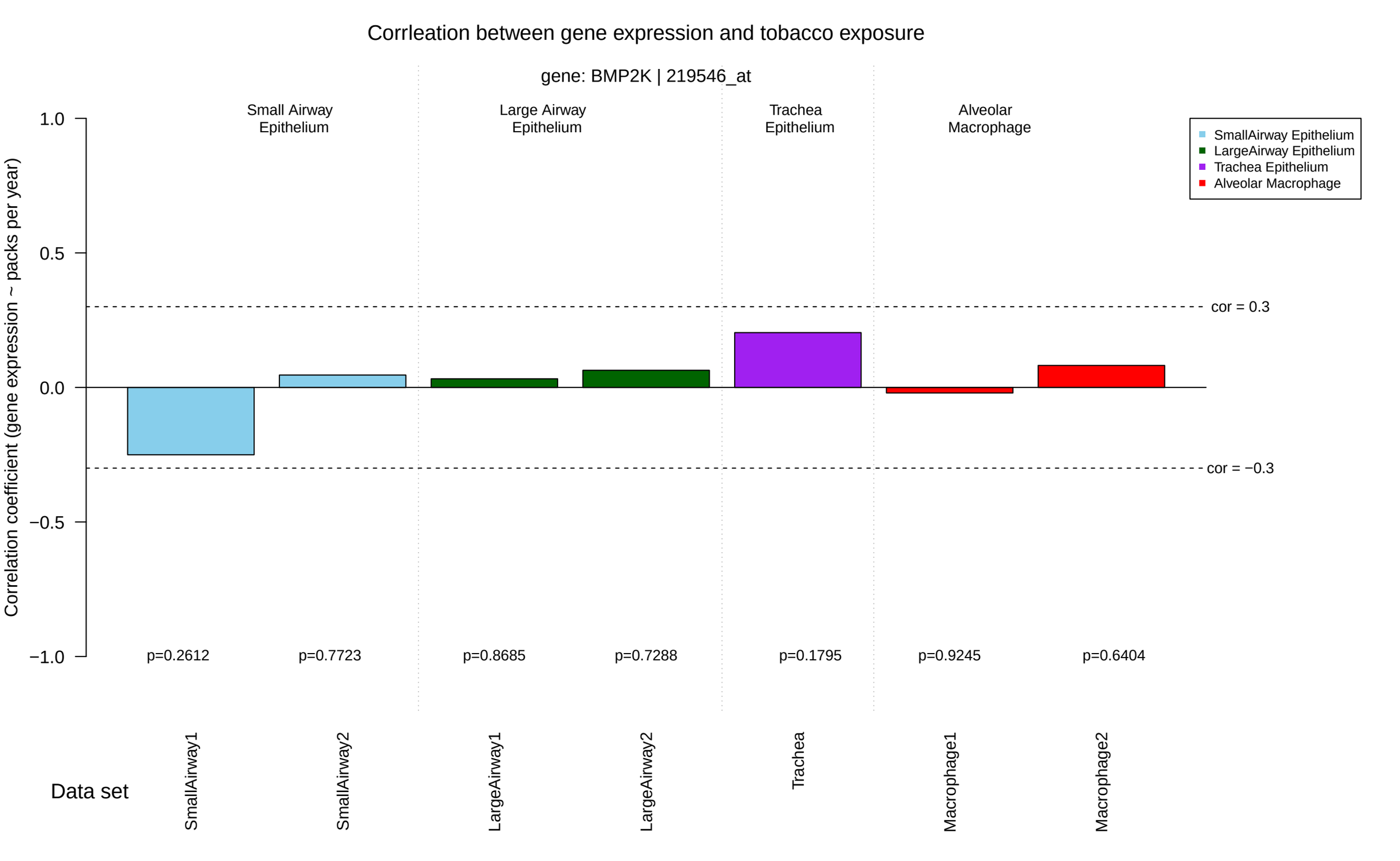

set: LargeAirway1 (p=0.9663)
BMP2K | 219546_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5969)
BMP2K | 219546_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7386)
BMP2K | 219546_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3408)
BMP2K | 219546_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7894)
BMP2K | 219546_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BMP2K | 219546_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2612
p=0.7723
p=0.8685
p=0.7288
p=0.1795
p=0.9245
p=0.6404
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2